Substitution

1 For each of the following equations, state whether the given value of x is the solution:

a $17 + x = 40, \quad x = 20$

b $32 + x = 58, \quad x = 26$

c $2x = 10, \quad x = 5$

d $\dfrac{x}{9} = 8, \quad x = 71$

e $\dfrac{x}{10} = 6, \quad x = 60$

f $8 - x = 10, \quad x = 2$

g $4x = -12, \quad x = -3$

h $x - 30 = 21, \quad x = 48$

i $x - 18 = 14, \quad x = 32$

j $7x = 33, \quad x = 5$

k $\dfrac{x}{4} = -4, \quad x = 16$

l $\dfrac{x}{20} = 5, \quad x = 100$

2 For each of the following, state whether the given value of the pronumeral is the solution of the equation:

a $u + 33 = 48, \quad u = 22$

b $v + 27 = 63, \quad v = 27$

c $t - 32 = 31, \quad t = 57$

d $48 - v = 11, \quad v = 37$

e $7r = 63, \quad r = 11$

f $\dfrac{v}{9} = 11, \quad v = 99$

3 Determine whether the following equations have $p = 8$ as a solution:

a $35 - p = 27$

b $\dfrac{50}{p} = 6$

c $6p = 48$

d $23 + p = 31$

e $26 - p = 23$

f $29 + p = 35$

g $10 + p = 28$

h $\dfrac{80}{p} = 10$

i $11p = 96$

j $p - 14 = 6$

k $p - 11 = -3$

l $24 + p = 34$

4 Determine whether the following equations have $r = 48$ as a solution:

a $2r = 93$

b $r + 12 = 60$

c $r - 15 = 33$

d $\dfrac{r}{6} = 8$

e $\dfrac{r}{6} = 5$

f $17 + r = 65$

g $3r = 134$

h $r + 25 = 73$

i $95 - r = 47$

j $\dfrac{r}{4} = 22$

k $\dfrac{96}{r} = 2$

l $28 + r = 76$

5 Determine whether the following equations have $y = 8$ as a solution:

a $5y = 40$

b $23 + y = 25$

c $22 - y = 18$

d $\dfrac{y}{4} = 2$

e $4y = 29$

f $45 - y = 32$

g $8y = 64$

h $y + 35 = 45$

i $56 - y = 38$

j $\dfrac{40}{y} = 5$

k $3y = 16$

l $y - 3 = 5$

6 Which of the following is the solution to the equation $x - 10 = 30$?

A 20 **B** 40 **C** 3 **D** 300

Trial and error

7 Consider the equation $56 - t = 39$.

a Which side of the equation is bigger when substituting $t = 10$?

b Should we try a smaller or larger number than 10 to solve the equation?

c Which side of the equation is bigger when substituting $t = 20$.

d Hence, what can be said about the solution of the equation?

8 Consider the equation $2u = 70$.

a Which side of the equation is bigger when substituting $u = 30$?

b Should we try a smaller or larger number than 30 to solve the equation?

c Which side of the equation is bigger when substituting $u = 40$?

d Hence, what can be said about the solution of the equation?

9 Consider the equation $m - 17 = 12$.

a Complete the table:

m	25	26	27	28	29	30
$m - 17$						

b Hence, what value of m will make the equation true?

10 Consider the equation $4k = 96$.

a Complete the table:

k	20	21	22	23	24	25
$4k$						

b Hence, what value of k will make the equation true?

11 Use trial and error to find the solution of each of the following equations:

a $47-t=14$ b $3v=72$ c $y+20=36$ d $x+24=43$

e $t+19=35$ f $s+12=38$ g $n-16=8$ h $5j=120$